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PALEOMAGNETISM OF THE BONIN ISLANDS AND ITS TECTONIC SIGNIFICANCE

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ABSTRACT

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Paleomagnetic samples were collected from six of the Bonin Islands located on the northeastern margin (27°N, 142°E) of the Philippine Sea. These volcanic islands are of Eocene age. Both lava flows and dikes were collected at each of 34 sites with Chichi-jima the largest island being sampled most extensively. Mean directions of the 27 sites on Chichi-jima are divisible into two groups showing antipolar relationships at the 95% confidence level. A between-site mean direction after tilt correction of 22 of the 27 sites is $I = 7°$, $D = 110°$, $\alpha_{95} = 14°$ while that of the remaining sites is $I = 10°$, $D = 273°$, $\alpha_{95} = 32°$. The seven sites on the other three Bonin Islands, as a whole, seem to cluster well around the mean of $I = 3°$, $D = 213°$ with $\alpha_{95} = 19°$, which differs significantly from both mean directions determined for Chichi-jima. The low inclinations and deflected declinations of the Bonin Islands are interpreted as being due to tectonic rotation. The islands have undergone a northward migration of at least 30° from the equatorial region together with (possibly clockwise) rotation of 30° to over 90° around the vertical axis. Since the northward drift and clockwise rotation are consistent with the tectonic movement inferred for the West Philippine Basin by previous studies, the Bonin Islands appear to have been part of the basin before opening of the Shikoku and Parece Vela basins during the Miocene.

INTRODUCTION

The Bonin, or Ogasawara Islands consist of several islands of volcanic origin located at about 27°N, 142°E on the Bonin Ridge (Island Arc). The ridge forms a part of the northeast margin of the Philippine Sea (Fig. 1). The eastern flank of the ridge grades gradually to the Izu-Bonin Trench, while the western side deepens steeply to the Bonin Trough a 3000–4000 m deep abyssal plain. At about 25°N, the Bonin Ridge and the Iwojima Ridge coalesce and become the Mariana island arc. The Mariana Trench extends southward in an arc convex toward the Pacific plate.

NE to NNE trending en-echelon ridges and troughs are pronounced in the Bonin Ridge and the Iwojima Ridge. Some of the structures extend completely across the

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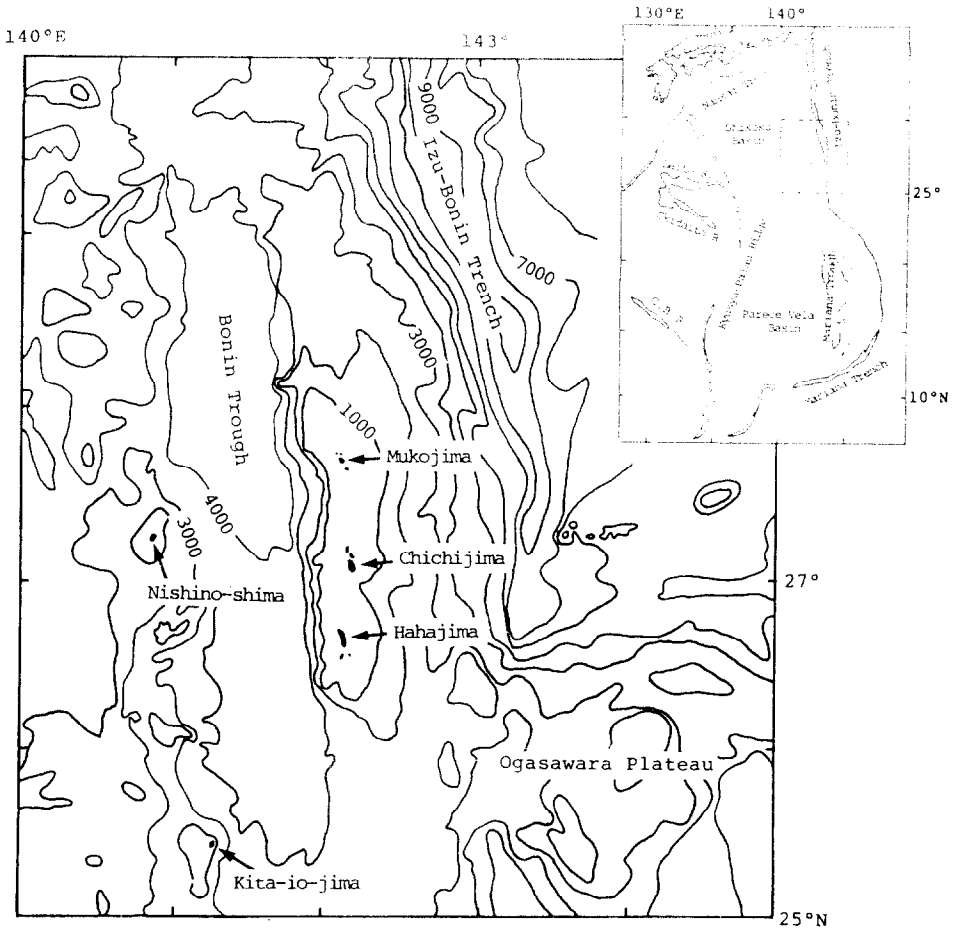


Fig. 1. Geomorphological outline of the Philippine Sea (inset) and regional bathymetric map around the Bonin Islands. *C.B.R.* = Central Basin Ridge (or Fault). Reproduced partly from the chart by the Hydrographic Office, Maritime Safety Agency of Japan (1966).

main part of the Bonin Ridge (Karig and Moore, 1975), and their trends appear to be separated by several fault zones (Bandy and Hilde, 1980).

The tectonic setting surrounding the Bonin Islands would indicate that these islands have undergone a significant movement, both on a local and regional scale. The en-echelon pattern of NNE ridges and troughs probably reflects the general tectonic disturbance of the Bonin Islands. The en-echelon structures may have produced shear movements along the Bonin Islands, rotating the islands along the vertical axis. As discussed in detail in a later section, the Bonin Islands are likely to have been formed during the Eocene prior to the opening of the Shikoku and Parece Vela basins. Previous studies (e.g., Louden, 1977) suggested that the nearby Philippine Sea plate drifted northward from the equatorial region and rotated clockwise by

several tens of degrees. Plate movements of the subducting Pacific plate could have produced additional displacements.

Paleomagnetic work in several of the Bonin Islands was begun in 1980 in order to examine the validity of the tectonic movements suggested in previous models. Sampling was conducted on three Bonin Islands, namely Chichi-jima, Ani-jima and Ototo-jima. The paleomagnetic results from Chichi-jima, the geologically best known of the Bonin Islands are presented here together with the preliminary paleomagnetic data for the other Bonin Islands (Kodama, 1981). Results for the islands, Ani-jima and Ototo-jima, will be published elsewhere in the near future.

GEOLOGICAL SETTING OF THE BONIN ISLANDS

The Bonin Islands consist of three groups of islands, each group lying about 50 km apart (Fig. 1). Each group is made up of a few islands and small islets up to some 20 km^2.

Since the end of the 19th century, the Bonin Islands have been studied by a number of scientists mainly from a petrological and paleontological point of view. Kikuchi (1890) first reported the characteristic pyroxene components in volcanic rocks on Chichi-jima. Peterson (1891) gave the name "boninite" to these andesitic rocks rich in bronzite phenocrysts. The rare volcanic rock called "boninite", a vesicular two pyroxene feldspar-free glassy andesite with a high content of MgO (6–12%) (Shiraki and Kuroda, 1977), occurs as the main constituent of the oldest volcanic events in the Bonin Islands. No volcanic rock exactly equivalent to the boninite has yet been found elsewhere (Shiraki and Kuroda, 1977), although similar rocks have been reported recently for the fore-arc region of the Marianas (Hussong et al., 1978). Early in this century paleontological studies of the *Nummulites* from Haha-jima were carried out by such workers as Yoshiwara (1902), Yabe (1920) and Hanzawa (1925). Since then several reports on the stratigraphy, topography, mineralogy and paleontology of the Bonin Islands have been published (e.g., Shiraki and Kuroda, 1977; Ujiie and Matsumaru, 1977). Detailed geologic descriptions covering the Bonin Islands, with the exception of Chichi-jima and Haha-jima, however, have never been completed. Very limited geologic information has been reported for islands such as Muko-jima, Ani-jima and Ototo-jima. In the following we summarize briefly the general geology of the two major islands from which most of our paleomagnetic data were obtained.

Chichi-jima

Chichi-jima comprises mainly pillow lava, volcanic breccia, andesitic dikes and subordinate limestone and sandstone (Fig. 2). The pillow lavas are either boninite or bronzite-andesite. The andesite pillows are commonly over three meters in diameter as contrasted with less than one meter for the boninitic pillows. Volcanic breccias

are composed of boninite, bronzite-andesite and dacite. The lower portions of the geologic section exposed consist generally of pillow lavas accompanied by a number of boninitic or andesitic dikes. Middle to upper horizons are composed of an alternation of pillow lavas and glassy hyaloclastites. Andesitic and partly dacitic massive lavas are distributed in the uppermost horizons and center of the island. The dip of these volcanic layers is gentle, generally less than 20°, except for a few locally complicated structures with dips of up to 60°. Dikes are common throughout the island. A swarm of tens of dikes showing the trend of NNW–SSE direction is exposed along the eastern and northern coast. The strike of this dike swarm changes to a WNW direction along the northern part of the island. A dike swarm with the same trend (WNW–ESE) can also be observed in the southwestern half of Ani-jima. Sedimentary rocks are distributed locally in the southwestern area of Chichi-jima and nearby small islets. The limestones exposed on the islets southwest of Chichi-jima appear to overlie unconformably the major volcanics of Chichi-jima. These limestones are reported to contain larger Foraminifera of Oligocene–Early Miocene age (Hanzawa, 1925; Iwasaki and Aoshima, 1970). Recently, much older fossil and absolute ages were obtained from samples from Chichi-jima; planktonic Foraminifera, Radiolaria and nannoplankton of Lower Eocene were discovered from the tuffaceous sediments in the northwestern part of the island (Y. Takayanagi, pers. commun., 1981), and K-Ar ages of about 40 Ma were reported for boninitic and dacitic rocks (Tsunakawa, 1983).

Haha-jima

Haha-jima, or Hillsborough Island, is the largest of the southernmost group of the Bonin Islands (Fig. 1). The occurrence of the Eocene fossils on this island is well known, however, no extensive geologic investigations have been performed. Brief studies by such workers as Hanzawa (1925), Iwasaki and Aoshima (1970) and Ujiie and Matsumaru (1977) have been reported. According to them, Haha-jima is composed mainly of andesitic (partly basaltic) lavas and volcanic breccias accompanied by a number of dikes. No pillow lavas are known from this island, in contrast with Chichi-jima where pillow lavas of boninitic compositions are dominant. Volcanic layers seem to be dislocated by several faults running NE or NNE in the island. Most of the strata are gently dipping, usually less than 20°, and show broad scale warping. Marine sedimentary layers of tuffaceous sandstone and sandy limestone are intercalated in the volcanics in the southern part of the island. These sediments contain various kinds of fossils such as larger Foraminifera (e.g., *Nummulites boninensis*) and planktonic Foraminifera, both indicating an age of late Middle Eocene (Saito, 1962; Ujiie and Matsumaru, 1977). Kaneoka et al. (1970) reported the K-Ar ages of 40 Ma (late Eocene) for a two-pyroxene andesite from the northernmost part of the island.

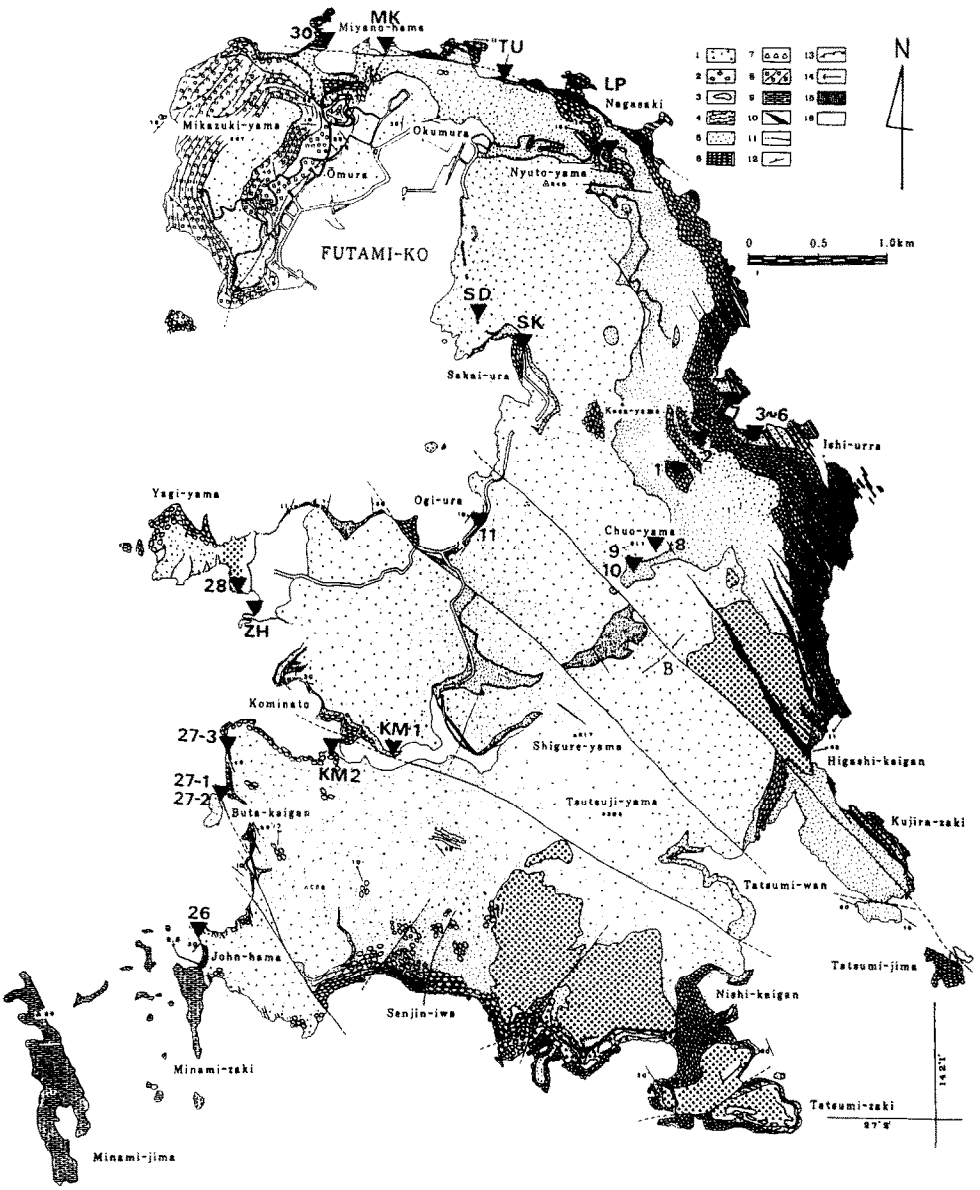


Fig. 2. Geological map of Chichi-jima after Maruyama and Kuramoto (1981), showing sampling localities (solid triangles) for paleomagnetic study. Sites indicated by numbers *1–30* were visited in this work, the others such as *LP*, *ZH* etc. were in Kodama (1981). Key: *1* = hyaloclastite of two pyroxene-plagioclase andesite, augite-plagioclase andesite and plagioclase andesite; *2* = pillow lava of two pyroxene-plagioclase andesite and augite-plagioclase andesite; *3* = homogeneous lava flow; *4* = reworked hyaloclastite; *5* = boninite hyaloclastite; *6* = boninite pillow lava; *7* = quartz-plagioclase rhyolite lava; *8* = conglomerate and sandstone; *9* = limestone; *10* = dike; *11* = fault; *12* = dip and strike, *13* = fault scarp by active landslide, *14* = direction of pillow tube, *15* = sandstone, *16* = beach sand.

Other islands

The remaining geologic information on the islands consists of either remote observations from boats or by short landing at several sites by the authors. Ani-jima and Ototo-jima are similar to Chichi-jima consisting mainly of boninitic or andesitic pillow lavas and hyaloclastites accompanied by a number of dikes. Pillow lavas are abundant in lower horizons and volcanic breccias or hyaloclastites are dominant in the upper horizons. Andesitic and dacitic massive lavas are distributed locally in the uppermost horizon. Several major faults with a NW–SE trend are present on Ani-jima while faults with a NNE–SSW trend are most common on Ototo-jima. Although the amount and sense of displacements of these faults remain obscure, the two sets of faults are likely to be conjugate (Maruyama and Kuramoto, 1981). The other small islets in the Chichi-jima group are also composed mainly of pillows lavas, hyaloclastites and dikes. Muko-jima, the largest island of the Muko-jima group, consists of pillow lavas of boninite and bronzite–andesite, hyaloclastites and many dikes in the same manner as Chichi-jima island (Kodama, 1981). The islands Yome-jima and Nakodo-jima, of Muko-jima islands group, consist of volcanoclastics and dikes of either boninitic or andesitic composition with lesser amount of pillow lavas.

SAMPLING AND LABORATORY PROCEDURES

Because of the steep topography of the island interiors and limited development of good exposures, our sampling sites were restricted to shorelines and a few interior roadside exposures. Sampling for paleomagnetic work on Chichi-jima, Haha-jima, Muko-jima and Yome-jima was made in June 1980 by one of us (K.K.) when he visited the four islands. Most of these samples were collected as separately oriented blocks from which one to three cores of 2.5 cm in diameter were drilled and cut into 2.5 cm cylindrical specimens. A total of 15 sites (118 specimens) were collected and the result of the reconnaissance work has been reported separately (Kodama, 1981). The next sampling was carried out in September 1980 by the present authors, focusing on the exposures on the Chichi-jima island group. All of the samples were cored in situ with the use of a portable rock drill from as many individual lava flows and dike units as time and exposures allowed. We define a paleomagnetic sampling site as being discrete in time and space, so that each flow or dike was considered to be a separate site if they were different from each other in stratigraphy or from far removed geographical locations that could not be stratigraphically correlated. In most cases one specimen was cut from the deepest part of each core in order to minimize the effect of surface weathering. In this article we present the new result from the 19 sites on Chichi-jima so far obtained at the second sampling, together with the summary of the first preliminary work on the several islands. The results from the remaining sites are discussed elsewhere (Keating, et al., 1983). Our

sampling focused on exposures of the major volcanics such as pillow lavas and dikes with boninitic or andesitic composition. Generally speaking, the boninitic or andesitic pillow lavas are fresh and have several centimeter thick glassy margins. The inner portions of the pillows are much more weathered than the margins in most cases. Therefore coring or block sampling focused on the fresh rims or near-rim portions of such pillows. Dikes are usually fresh and homogeneous, frequently displaying well developed columnar joints. A few samples were collected from coarse-grained hyaloclastites on Chichi-jima, but they were found to be magnetized very weakly (less than 10^{-4} A/m) and represent unstable behaviors. Thus these results are omitted from statistical and tectonic analyses. Where layers of sedimentary rocks overlying or underlying the sampling sites were clearly identified, strike and dip of the strata were used for structural correction for the in-situ magnetic directions. Although the amount of primary vs. secondary tilting was in some cases uncertain, we attempted to make bedding corrections for about one third of the sites on Chichi-jima where nearby bedded layers were recognizable. At no other sites could we find clear evidence of structural dislocation such as tilted sedimentary layers or pillows accompanied with remarkably inclined long axis. Almost all of the pillows preserved massive shapes suggesting their solidification on nearly horizontal planes rather than an inclined surface.

Remanent magnetizations of all the specimens were measured using a Schonsted SSM-1A spinner magnetometer at the Paleomagnetic Laboratory of the Geophysical Institute, University of Tokyo. Following the initial measurement of natural remanent magnetization (NRM) of all samples, several pilot specimens were chosen from each site and remeasured after progressive demagnetization in alternating fields with peak intensities from fifty to several hundred oersted (Oe). Then the optimum demagnetization field for the remaining specimens was selected as that above which systematic change in direction seemingly ceased. In addition to AF demagnetization, several representative specimens from separated sites were demagnetized thermally from 150° to 550°C in nitrogen gas. The magnetic stabilities of representative samples are illustrated in orthogonal plots of the magnetization vectors during AF demagnetization in Fig. 3. It is clear that the magnetization of pillow lavas is fairly stable up to the range of 800–1000 Oe in peak field. These samples are characterized by high magnetic coercivity with median destructive field (MDF) values of 200–300 Oe. As a result of these pilot lava studies, the remaining samples of each site were demagnetized in fields of 100–150 Oe in order to eliminate any soft components of magnetization. Although a few specimens which appeared weathered more deeply were contaminated by secondary components as, for example, in the specimen NS 101 in Fig. 3, such magnetic overprint could be erased easily by relatively low fields of 200 and 250 Oe. The observations of stable magnetic remanence were reinforced by the results of thermal demagnetization studies as shown in Fig. 4. No remarkable change of direction was found up to over 500°C. The gradual decrease in total intensity during heating suggests high blocking temperature of magnetic carriers.

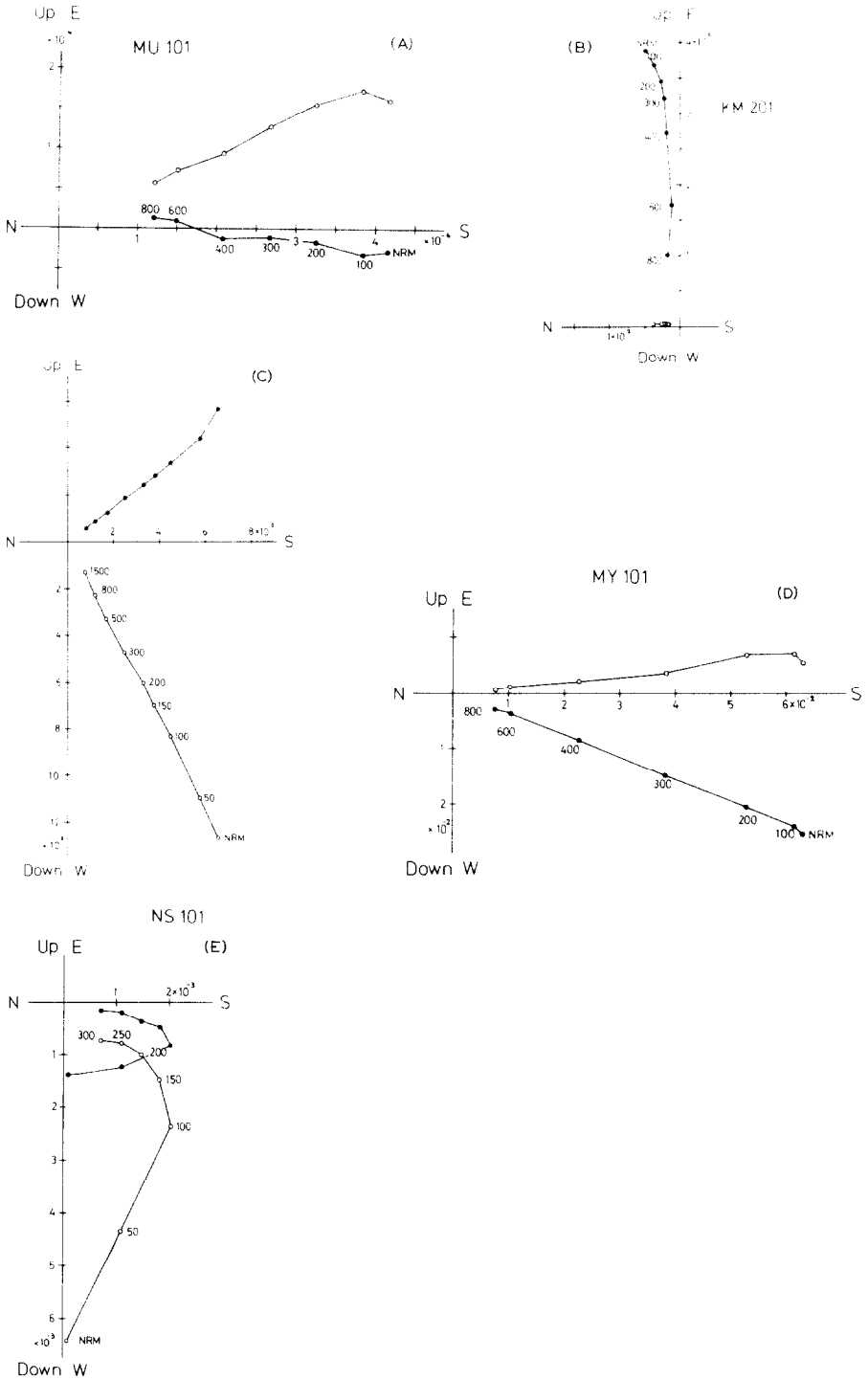


Fig. 3. Orthogonal plots of successive endpoints of magnetization vector during progressive AF demagnetization. Numbers are intensities of peak fields in Oe. Solid circles represent projections on the horizontal plane, and open circles those on the north–south vertical plane. Unit in emu. A, B, C. Boninite pillows at site MU-1 (Muko-jima), KM and 27-3 (Chichi-jima). D, E. Andesite lavas at site MY and NS (Haha-jima).

Thermomagnetic analysis in Fig. 5, on the other hand, demonstrated the irreversible nature for some representative specimens (boninite pillow and andesite lava), that is, the heating curves revealed high and low Curie temperatures while the cooling curves gave a single high Curie point. We believe that these results are indicative of the presence of thermally unstable titanomaghemite produced possibly by subaqueous low temperature oxidation as typically observed in abyssal basalts (Ozima and Larson, 1970). Although evidence of alteration in magnetic minerals could be more or less recognized by the thermomagnetic examination, the stability of magnetization

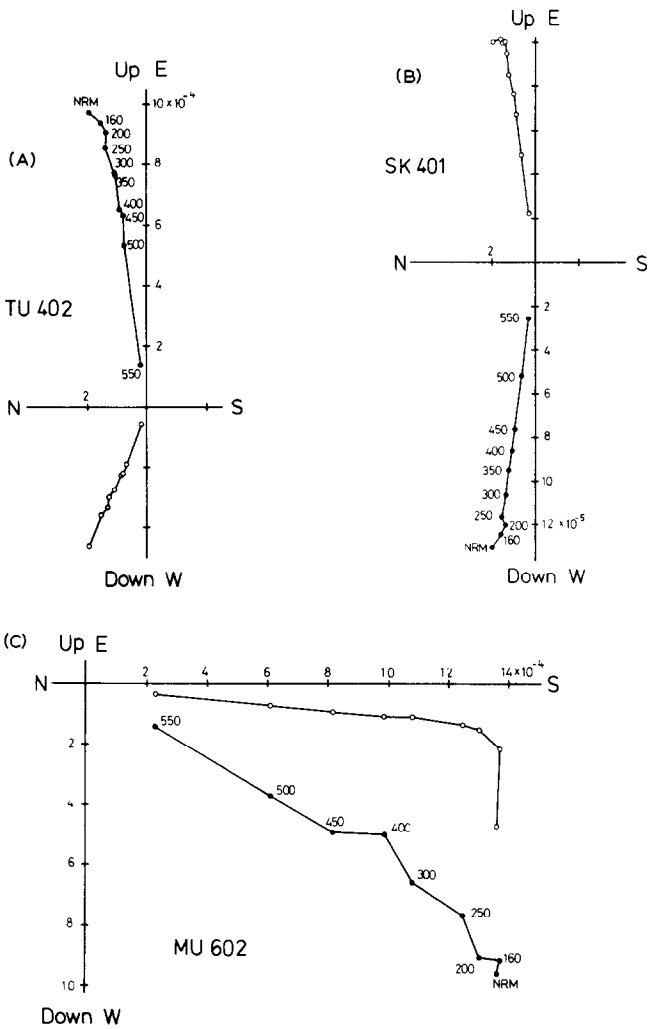


Fig. 4. Orthogonal projections of magnetization vector through progressive thermal demagnetization. Numbers indicate demagnetization temperatures in centigrade degrees. Symbols same as in Fig. 3. A, B. Boninite pillows at site TU, SK on Chichi-jima. C. Andesite lava at site MY on Haha-jima.

during progressive AF and thermal demagnetization treatments means that most of the remanence measured is due to the initial thermo-remanent magnetization (TRM) free from secondary overprint such as partial TRM (PTRM) due to reheating. In contrast to the overall magnetic stability of both pillow lavas and dikes, their NRM intensities were different on average by more than one order of magnitude; that is, those of pillow lavas were 10^{-1}–10^{-2} A/m those for andesite or dacite dikes 1–10 A/m. The weaker intensity of the pillow lavas, both boninite and bronzite-andesite, could possibly be attributed to high vesicularity of 10–30% (Shiraki and Kuroda, 1977) and high percentage of glassy materials, resulting in the smaller amount of magnetic minerals.

All of the within-site mean magnetic directions in Table I are plotted in Fig. 6, where data from Chichi-jima and the other Bonin Islands are shown separately for the convenience of interpretation. The mean directions throughout the Bonin Islands are deflected significantly from the normal or reverse direction expected for the present position of the islands. Based upon the limited number of sites reported here, it appears that the mean directions are distinctly different, especially in declinations, between Chichi-jima and the other islands. The site means of Chichi-jima are divisible apparently into two opposite polarities; that is, the major populations of the means tend to cluster into one group with easterly deflected declinations while the remainder, despite their larger dispersion as a whole, seem to be displaced westerly. Giving an equal weight to each site mean direction, the 22 sites with

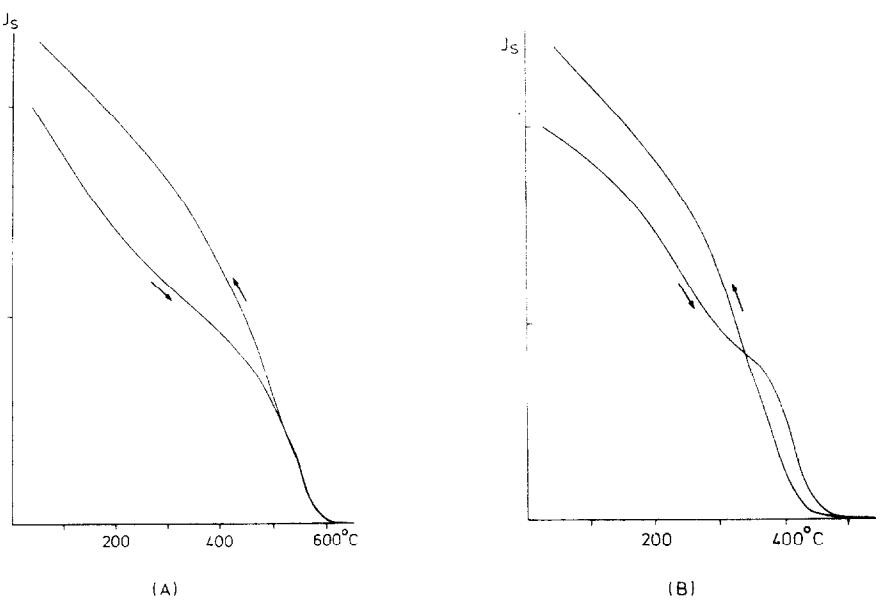


Fig. 5. Thermomagnetic curves of boninite pillow (A) at site KM (Chichi-jima) and andesite lava (B) at site MY (Haha-jima). Vertical axis is relative intensity of saturation magnetization.

TABLE I

Summary of paleomagnetic data from the Bonin Islands *

Site	Rock type	N	I	D	k	α_{95}	H
Chichi-jima							
1	BP	14	13.0	139.8	79.8	4.4	100
2	BP	8	−43.9	127.8	415.7	2.7	100
			(−10.3)	(108.3)			
3	BP	7	12.8	108.2	216.0	4.1	150
4-1	AD	9	7.4	137.6	157.3	4.1	200
4-2	AD	7	−53.8	167.2	357.1	3.2	150
5	BP	7	2.4	114.6	267.6	3.7	100
6	AD	4	16.0	252.7	295.2	5.4	150
7	BP	9	13.7	97.5	270.5	3.1	150
8	DL	14	30.3	125.3	29.5	7.5	200
			(42.0)	(132.2)			
9	AL	12	25.5	140.4	199.2	3.1	100
			(34.4)	(147.4)			
10	DD	4	15.3	132.5	2725.3	1.8	100
			(26.0)	(136.5)			
11	BD	6	39.9	284.3	438.0	3.2	150
			(15.5)	(281.3)			
26	AD	15	2.9	110.9	24.3	7.9	100
			(−0.7)	(111.1)			
27-1	AD	6	−2.7	110.0	45.6	10.0	200
27-2	AD	4	−31.9	84.8	685.3	3.5	200
27-3	AP	13	60.3	149.5	424.6	2.1	100
			(20.4)	(105.6)			
28	DL	14	−7.5	299.1	38.7	6.5	150
30-1	BP	8	24.5	79.5	52.0	7.8	150
			(23.2)	(103.3)			
30-2	BP	11	1.3	88.6	133.0	4.1	100
			(1.9)	(90.1)			
LP	AP	13	−53.7	87.9	70.7	5.1	150
ZH	DD	5	47.9	125.8	149.0	6.3	100
KM-1	BP	7	−7.2	88.2	309.8	3.6	100
KM-2	DL	7	−15.2	90.6	42.6	9.2	100
MK	BP	6	31.0	75.4	52.6	9.2	250
TU	BP	7	27.4	90.4	138.0	5.1	250
SD	DD	5	−41.5	261.8	243.0	5.1	150
SK	BP	9	−32.5	266.5	13.3	14.8	250
Haha-jima							
NK	AL	12	−10.2	210.7	77.0	5.1	100
KK	AL	6	22.1	234.9	23.6	13.8	150
NS	AL	13	8.4	190.7	23.1	8.9	250
MY	AL	11	−2.2	209.7	67.1	5.7	100
Muko-jima							
MU-1	BP	8	−25.0	203.7	31.8	9.9	100

TABLE 1 (continued)

Site	Rock type	N	I	D	k	α_{95}	H
MU-2	BP	5	14.1	209.2	663.1	2.6	100
Yome-jima							
YM	AD	4	−26.8	232.8	7.4	37.4	100

* Rock type: AD = andesite dike, AL = andesite massive lava, AP = andesite pillow lava, BD = boninite dike, BP = boninite pillow lava, DD = dacite dike, DL = dacite massive lava. N: number of specimens. I, D—inclination and declination in degrees (those after correction for bedding tilt are shown in parenthesis), k—Fisher's precision parameter, α_{95}—semiangle of 95% confidence cone, H—peak intensity in Oersted used for AF demagnetization. Data from the lower fifteen sites (site LP to YM) are after Kodama (1981).

easterly declinations give an overall mean direction of $I = 6.0°$, $D = 110.5°$ with a 95% confidence radius (α_{95}) of 15.8° whereas the remaining five sites give that of $I = -5.7°$, $D = 273.2°$ with $\alpha_{95} = 40.0°$. Apart from the large scatter of the latter group, the two means are of antipolar relation at the 95% confidence level. This may be regarded as additional evidence, independent of the laboratory stability tests, that the remanences measured are primary magnetizations free of significant secondary overprint such as chemical magnetization due, for example, to titanomaghemitization. Furthermore, the presence of the apparent normal and reversed polarities excludes the possibility that the displaced directions may be due to some anomalous paleogeomagnetic behavior such as excursions during polarity transitions.

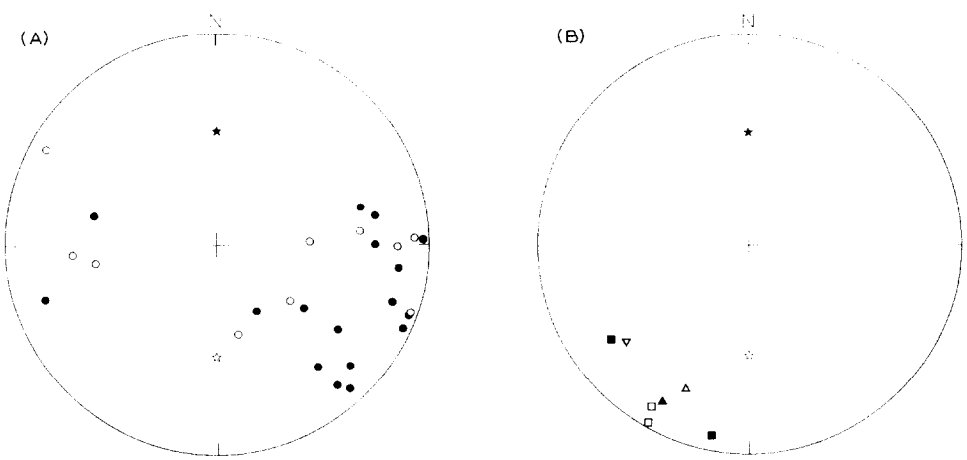


Fig. 6. All of the in-situ mean remanence directions of Chichi-jima (A) and the other three islands (B). Solid (open) symbols plots on the lower (upper) hemisphere. ● = Chichi-jima, ■ = Haha-jima, ▲ = Muko-jima, ▼ = Yome-jima. Star represents direction of axial dipole field expected at the present latitude of the Bonin Islands.

It may be possible, on the other hand, that some parts of the angular dispersion of the mean directions in Chichi-jima are attributed to unresolvable local structural disturbances in addition to geomagnetic secular variation. Structural corrections were carried out for nine sites in Chichi-jima, the corrected data and bedding parameters being listed in Table I. Although only one third of the sites were selected for structural correction by using the strike and dip of interflow sedimentary layers or layers directly overlying or underlying each exposure, a little better grouping of between-site mean directions was obtained by such correction. The average direction of the 22 sites with easterly declinations, if it includes the eight sites thus corrected, becomes $I = 7.0°$, $D = 110.1°$ and $\alpha_{95} = 14.3°$, while that of the antipolar group including one corrected site (site 11 in Table I) is $I = -10.3°$, $D = 273.0°$ and $\alpha_{95} = 32.2°$. It is notable also in this case that those two groups after the structural corrections remain antipolar with each other on the 95% confidence level. If the westerly directions of the five sites are inverted through the origin, an overall mean direction of the 27 sites in Chichi-jima becomes $I = 7.7°$, $D = 107.7°$ with $\alpha_{95} = 12.5°$. Although the true mean direction which would be obtained after the necessary structural correction for all the sites would have smaller error radius, it is of great importance that this estimate of the mean even after the partial structural correction is significantly displaced from the axial dipole field direction at the present latitude of Chichi-jima. The sites on the other Bonin Islands, even more interestingly, indicate a fairly different direction. The seven mean directions of the three other islands tend to cluster around an overall mean of $I = -2.8°$, $D = 212.7°$ with $\alpha_{95} = 10.0°$. Although this estimate may be less reliable than that of Chichi-jima because of the small number of sampling sites and poor geological information for each island, the relatively good grouping seems to indicate a feature of importance. Since the two exposures on Haha-jima (site NK and MY) are massive andesite lava flows intercalated with distinct marine sedimentary layers which are bedded nearly horizontally, the uncorrected data of the two are believed to represent the original (structurally undisturbed) directions, and therefore may be most reliable. The other five sites have no structural correction data and therefore were not corrected but given equal weight to calculate an overall mean direction.

Summarizing the paleomagnetic results described above, the Bonin Islands can be characterized by (1) the significantly different declinations between Chichi-jima and the other islands, (2) the consistent low inclinations (around zero) throughout all the islands and (3) the occurrence of normal and reversed polarities on Chichi-jima. The first feature suggests strongly that the individual Bonin Islands have undergone more or less specific rotations, clockwise or counter-clockwise, around the vertical axis, although the sense and amount of the rotation differ significantly between Chichi-jima and the other islands. The second feature means that the Bonin Islands as a whole had been formed far south of their present position, possibly near the equatorial region, and then drifted northward.

DISCUSSIONS

Because the declinations are deflected remarkably to the east or west, it is difficult to identify which group of field directions should be assigned to a normal or reversed polarity epoch. If a large number of the sites with easterly declinations in Chichi-jima are assigned to a normal epoch, then the results can be interpreted such that the island had been formed in the northern hemisphere and undergone large clockwise rotation of over 90°. If, alternatively, the easterly deflected group is assigned to the reversed polarity, it follows that Chichi-jima originated in the southern hemisphere and then rotated less than 90° in an counter-clockwise sense. Although this ambiguity cannot be resolved by the paleomagnetic result restricted to Chichi-jima, additional data from the other Bonin Islands may be helpful to determine the sense of rotation. Noting that the mean direction of the seven sites ($I = 2.8°$, $D = 212.7°$) is displaced to the southwest less than that for Chichi-jima, it may be reasonable to assign the direction to a reversed polarity. If so, a proper interpretation is that the islands had been formed in the northern hemisphere very close to the equator and then underwent clockwise rotation together with northward migration. Assuming that the sense of rotation is the same among the Bonin Islands, Chichi-jima may also have undergone a much larger clockwise rotation ($\sim 100°$) than the other islands. The problem arising from this inference is why such differential rotations occurred to the Bonin Islands. One possible reason would be related to the regional geologic structure in and around the Bonin Islands. Tectonic complexities such as en-echelon ridges and troughs may be one of the most plausible reasons for the rotational movements around the vertical axis. A series of NNE–SSW trending strike-slip faults diagonal to the Bonin Ridge would cause various amounts of rotation to the blocks between them. Such rotational movements may be analogous to those of ball bearing as schematically shown in Fig. 7. The differential rotations between Chichi-jima and other islands would thus correspond to the differential motion of the strike-slip faults between them. Apart from the rotational movements, the northward drift of the Bonin Islands as demonstrated by our paleomagnetic data might be caused partly by the transportation associated with the northward oblique subduction of the Pacific plate beneath the Philippine Sea plate (Fig. 7).

In reviewing the previous studies on the tectonic development of the Philippine Sea, it is important to note that the present paleomagnetic data of the Bonin Islands is consistent with the tectonic motions demonstrated by analysis of marine magnetic anomalies and paleomagnetic data of DSDP sediments and Miocene volcanics on Guam. Karig (1975) suggests that the oldest part of the West Philippine Basin (WPB) is Middle–Late Eocene and that it has undergone both northward migration and clockwise rotation. Shih (1978) uses a phase shifting technique in the analysis of the marine magnetic lineations from the WPB and concludes that the basin drifted northward by about 20–30° and rotated clockwise about 50–60° since 35–40 Ma.

The northward drift and clockwise rotation of the basin was proposed also by Louden (1977), based on a paleomagnetic study of DSDP sediments and detailed analysis of several marine magnetic anomalies. The northward motion was sup-

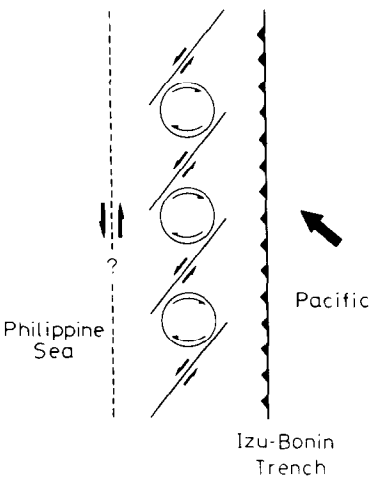


Fig. 7. A possible model allowing tectonic rotation about vertical axis, accompanied by lateral transportation of blocks in strike-slip fault zones. Modified and combined from models by Fitch (1972) and Beck (1980).

ported recently by paleomagnetic study on the DSDP cores recovered from the West Philippine Basin (Kinoshita, 1980; Keating 1981) which indicates a linear increase of paleolatitude as a function of age from near the equator to the present latitude. In addition to the investigations of the oceanic areas, paleomagnetic work, on the Miocene rocks of Guam (Kobayashi, 1972; Larson et al., 1975) indicated that Guam underwent clockwise rotation of 50–60° with little, if any, latitudinal drift. Recent paleomagnetic data by Keating et al. (1983) also suggests large rotations for several Mariana Islands including Guam, Saipan, Tinian and Rota. Figure 8 plots the paleomagnetic pole positions obtained by the previous studies including the present result for the Bonin Islands. In the figure the mean paleopole position of Chichi-jima and other islands was estimated by assigning the easterly directions to a normal polarity, and by inverting the reversely magnetized directions. In spite of the large dispersion, paleopoles for the Bonin and the West Philippine Basin seem to be consistent with each other in the sense that they are displaced significantly to the far side from the north pole and indicate clockwise rotations as well.

Summarizing the discussions above, there are two main factors causing the tectonic movements to the Bonin Islands. One is related to the more local structures around the Bonin, represented by en-echelon structures. Another is the broad-scale tectonics covering the whole of the West Philippine Basin. Considering that the age

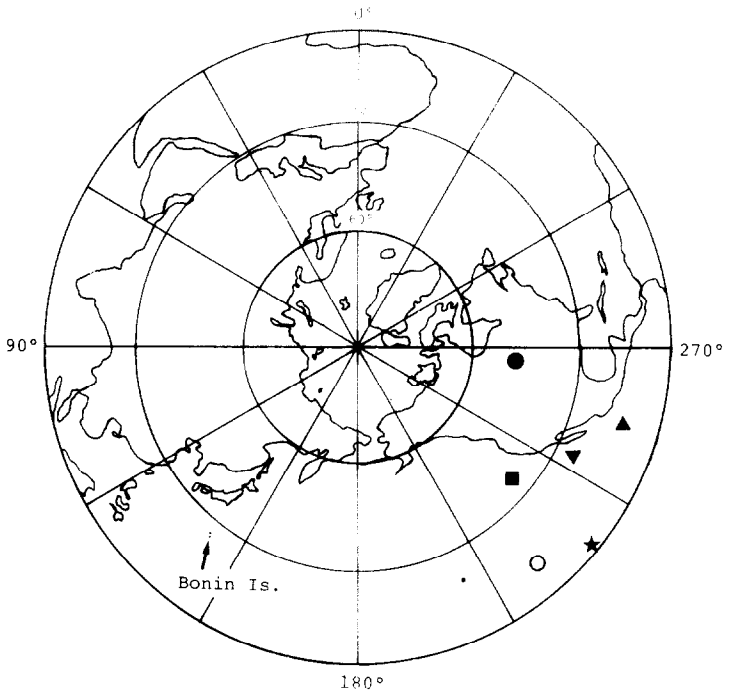


Fig. 8. Equal-area projections of paleomagnetic pole positions determined for various parts of the Philippine Sea. ★ (3°N, 130°W)—West Philippine Basin (Louden, 1977), ▼ (23°N, 117°W), ▲ (14°N, 106°W)—West Philippine Basin (Shih, 1978), ■ (37°N, 131°W)—Guam (Larson et al., 1975), ○ (14°S, 140°W)—Chichi-jima (this work), ● (49°N, 94°W)—three Bonin Islands (this work).

of the Bonin Islands (Eocene) is compared to that for the older part of the basin, it is possible that the islands had developed tectonically in harmony with the Philippine Sea, namely they underwent northward migration and clockwise rotation. We suggest that the Bonin Ridge was marginal to the West Philippine Basin and was formed by volcanic activity during the early history of the Philippine Sea prior to the formation of both Shikoku and Parece Vela basins. The fact that the larger Foraminifera of *Nummulites boninensis*, previously found only in one of the Bonin Islands (Haha-jima) was dredged from the Amami Plateau and the Daito Ridge (Mizuno et al., 1975; Research members of the GDP-11 cruise, 1975), 1000 km west of the Bonin Islanda across the Shikoku Basin, suggests that the Bonin Ridge should have been located at nearly the same position as the present northern part of the West Philippine Basin before opening of the Shikoku Basin. Finally, we would like to emphasize that, although it is still obscure which tectonics, regional or broad scale, was dominant as an agency of the tectonic movement of the Bonin Islands, extensive paleomagnetic study of the islands lying along the eastern rim of the Philippine Sea can give information useful to the resolution of this problem.

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